A2 Variables, Types, Expressions, and Control Structures

Modern Code
MC (MTD.ba VZ)
SS 2025

Submission via Moodle **24 Points**

This assignment focuses on variables, data types, expressions, and control structures in Java. You will create programs that demonstrate proper variable usage, arithmetic operations, and decision-making logic.

Submission Requirements:

- Read the `Organizational Guidelines on Moodle`, and adhere to them!
- Submit a PDF document containing your solution idea, code, and tests for each exercise
- Include all Java source files (.java) in a ZIP archive
- As tests include copies or screenshots of the console output
- Use the naming convention: `MC_UE02_LASTNAME.pdf` and `MC_UE02_LASTNAME.zip`

Important: For exercises 2 and 3, think about whether `if/else` or `switch` would be the better choice for the problem. For exercises 4-6, consider which loop type (`for`, `while`, or `do-while`) is most appropriate for the task.

Exercise 1 (4 points) Rectangle Calculator

Create a Java program called `RectangleCalculator.java` that calculates various properties of a rectangle. The program should:

- Declare variables for width and height (both `int`)
- Calculate and display: area, perimeter, and diagonal length
- Use proper variable names and include comments
- Test with different values for width and height
- Display results with proper formatting (use commas for decimal places)

Mathematical Functions:

- **Math.sqrt():** Returns the square root of a value
- **Example:** `double square_root = Math.sqrt(25);`

Expected Output:

```
=== Rectangle Calculator ===
Width: 5
Height: 3
Area: 15
Perimeter: 16
Diagonal length: 5.83
```

Exercise 2 (4 points) FizzBuzz Game

Create a Java program called `FizzBuzz.java` that implements the classic FizzBuzz game. The program should:

- Use a variable to store the maximum number to count to
- Check divisibility by 3 and 5
- Display ”Fizz” for numbers divisible by 3, ”Buzz” for numbers divisible by 5, ”FizzBuzz” for numbers divisible by both, and the number itself otherwise
- Use proper variable names and include comments

Expected Output:

```
=== FizzBuzz Game ===
Counting from 1 to 15:
1
2
Fizz
4
Buzz
Fizz
7
8
Fizz
```

```
Buzz
11
Fizz
13
14
FizzBuzz
```

Exercise 3 (4 points) Day of Week Processor

Create a Java program called `DayOfWeek.java` that processes days of the week. The program should:

- Use a variable to store a day number (1-7)
- Display the day name and type
- Display whether it's a weekday or weekend
- Handle invalid input appropriately
- Test with different day numbers
- Use proper variable names and include comments

Expected Output:

```
=== Day of Week Processor ===
Day number: 3
Day: Wednesday
Type: Weekday
```

Exercise 4 (4 points) Star Pattern Generator

Create a Java program called `StarPattern.java` that generates a star pattern. The program should:

- Use a variable to control the pattern size (number of rows)
- Display a right triangle pattern of stars
- Use proper variable names and include comments
- Test with different pattern sizes

Expected Output:

```
=== Star Pattern Generator ===
Pattern size: 5
*
* *
* * *
* * * *
* * * * *
```

Exercise 5 (4 points) Collatz Conjecture

Create a Java program called `CollatzConjecture.java` that explores the famous Collatz Conjecture. The program should:

- Start with a given number
- If the number is even, divide it by 2
- If the number is odd, multiply by 3 and add 1
- Repeat until you reach 1
- Display each step and count the total steps
- Use proper variable names and include comments

Expected Output:

```
=== Collatz Conjecture ===
Starting with 5:
5 (odd) -> 16
16 (even) -> 8
8 (even) -> 4
4 (even) -> 2
2 (even) -> 1
Reached 1 in 5 steps!
```

Exercise 6 (4 points) Dice Rolling Simulator

Create a Java program called `DiceRollingSimulator.java` that simulates rolling dice until you get a specific number. The program should:

- Use a variable for the target number (1-6)
- Roll dice until you get the target number
- Count how many rolls it takes
- Display each roll and the final count
- Use proper variable names and include comments

How to generate random numbers:

- Use `Math.random()` which returns a decimal between 0.0 and 1.0
- Multiply by 6 and add 1: `(int)(Math.random() * 6) + 1`
- This gives you a random integer from 1 to 6

Expected Output:

```
=== Dice Rolling Simulator ===
Target number: 6
Roll 1: 3
Roll 2: 1
Roll 3: 4
Roll 4: 6
Found target number 6 in 4 rolls!
```